

National Entrepreneurship Challenge (NEC)

E-Cell, IIT Bombay

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Team: Titan - Circular Economy & E-Waste

My Motivation

A site visit to informal e-waste yards in Seelampur was a wake-up call for me. Circuit boards were being burned in open fires and soaked in crude acid baths to extract tiny amounts of gold and copper, dumping toxic effluents directly into local drainage.

As a metallurgical engineering student, I know we have clean, closed-loop hydrometallurgical processes that can recover over 95% of precious metals safely. My goal is to build decentralized, containerized e-waste processing modules that operate cleanly near urban collection centers. Turning electronic waste into a safe, urban mining industry is the venture I want to build.

My Expectations from E-Cell

E-waste recycling involves strict Extended Producer Responsibility (EPR) mandates, CPCB authorization, and complex logistics. I expect E-Cell to connect our team with environmental policy advisors and industrial waste management veterans.

Through NEC, I want to figure out how to secure steady e-waste supply streams from corporate IT parks, calculate operational breakeven points for mini-refineries, and present our environmental compliance framework to impact investors.